MARKET-READINESS ROADMAP · V0.4.0 TO REVENUE-READY

LACUNA·AMR

The step-by-step path from the verified v0.4.0 audit platform to a market-ready AMR trust layer: current-state inventory, market positioning, the Vecura integration, six delivery phases with exit criteria, KPIs, risks and this week's actions.

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Open core (Apache-2.0) · Research Use Only
Audit & trust layer for AMR prediction

NOT FOR CLINICAL DECISION-MAKING

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1. Executive Summary

LACUNA-AMR is an open-core (Apache-2.0) audit and trust layer for antimicrobial resistance (AMR) prediction. Instead of adding another resistance predictor to a crowded field, it answers the question every laboratory, public-health agency and pharmaceutical AMR program eventually faces: **can this dataset and this model be trusted?** The platform ships five capabilities - dataset auditing, conformal prediction with statistically guaranteed uncertainty, split-leakage validation, publication-compliance reporting and reproducibility manifests - behind a web console, a versioned REST API, and an agent-native MCP tool server.

As of version 0.4.0 (9 November 2026) the platform is functionally complete for research use: a fourteen-section console, an enforced and metered **/api/v1** surface with OpenAPI 3.1, a JSON-RPC 2.0 MCP server exposing the five capabilities as agent-invocable tools, HMAC-signed webhooks, a hash-chained audit trail, and a Python package (v0.2.0, 60 tests) backed by a real exploratory pilot on 240 Escherichia coli genomes. All quality gates G0-G11 have been executed with evidence files, including two honestly documented failures (clonal leakage in sealed splits; fail-closed phase B) that were corrected rather than hidden.

This roadmap states the exact steps from that current state to a market-ready product. Market-ready is defined by five exit criteria: at least one pilot reference laboratory using the audit console weekly; self-serve API signup to first successful audit call in under 15 minutes; zero integrity-rule violations in shipped copy; a completed end-to-end MCP round-trip in joint testing with the Vecura platform; and a defensible regulatory posture (research-use-only claims, documented EU AI Act and MDR pathway). Six phases deliver those criteria in approximately twelve months.

> **The market window**
>
> Three independent 2025-2026 estimates place the AMR surveillance market at USD 6.56-6.67 billion in 2025, growing to USD 11.17-12.18 billion by 2034-2035 (CAGR 5.5-6.9%). WHO GLASS now aggregates data from more than 100 countries and over 23 million confirmed infections, while reporting persistent data-quality and comparability gaps - precisely the problem LACUNA-AMR audits. No incumbent sells an independent, license-clean dataset and model audit layer; the instrument vendors (bioMerieux, BD, Thermo Fisher) sell closed ecosystems, and the prediction tools (ResFinder, Mykrobe, DeepARG) ship point calls without leakage quantification or uncertainty guarantees.

2. Current State: Verified Assets and Honest Gaps

The project inventory below was verified on 2026-11-09 against the repository, the executed gate evidence and live endpoint tests. It separates what is real and tested from what is deliberately not built yet - a distinction that is itself a product feature in a market where overclaiming is endemic.

Asset	State	Evidence
Python package lacuna-amr v0.2.0	Shipped. CLI: audit-dataset, train, predict, validate, report, manifest; offline-enforced; 60 tests.	download/lacuna-amr; release archives; SBOM (SPDX 2.3)

Asset	State	Evidence
Research pilot (E. coli, N=240)	Executed, EXPLORATORY label. Ciprofloxacin AUROC 0.980 [0.82, 1.00]; gentamicin 0.915; ceftazidime 0.653 (S0, M1, cluster-bootstrap 95% CI).	results/evaluation/main_results.json; manuscript regenerated after D-019 correction
Quality gates G0-G11	All executed with evidence files. G3 (clonal leakage) and G5 (fail-closed phase B) documented as FAIL and handled honestly; v0.2.0 ships clone-aware splits.	gates/G0.md - G11.md; FINAL_REPORT.md
Web console v0.4.0	14 sections: overview, workflow, cohort, audit, predict, batch, validate, gates, provenance, market, services, integrations, compliance, report.	Live console; browser-verified 2026-11-09
Versioned API /api/v1	Enforced (401/402/403), metered (credits), scoped (read / read+predict), CORS-enabled. OpenAPI 3.1 at /api/openapi.json.	Web-layer gates W1-W2 PASS (gates/evidence/)
MCP server /api/mcp	JSON-RPC 2.0: initialize, tools/list (5 tools), tools/call; same auth and metering as REST.	W3 PASS
Vecura-ready webhooks	HMAC-SHA256 signed outbound events with honest delivery logging; inbound receiver with timing-safe verification.	W4-W5 PASS; self-loop delivery demonstrated
Compliance layer	Hash-chained append-only audit trail with one-click verifier; FHIR R4 and GLASS-style exports (labelled, no false format claims); RUO notice on every predictive surface.	D-025, D-027, D-028

Table 1. Verified asset inventory, checked 2026-11-09.

2.1 Honest gaps (deliberately not built yet)

Gap	Why it is open	Closes in
Multi-tenant authentication	Console is single-workspace; API keys are the only programmatic identity.	Phase 1
Payment processing	Entitlement and credit system is real; Stripe integration is stubbed pending legal entity.	Phase 1
Production database	SQLite development database; Postgres migration path documented.	Phase 1
Full-scale DG cohort	Pilot is 240 isolates against a data gate requiring at least 800 per drug; full-scale run specified but not executed (hardware-bound).	Phase 2
External benchmark	No head-to-head published comparison against ResFinder/Mykrobe-class tools on the audit task.	Phase 2
Regulatory clearance	Strictly research-use-only; no diagnostic claims anywhere.	Phase 3 (posture), optional SaMD path later
Vecura joint test	Connector is standards-based and self-loop tested; no joint test with the actual platform has occurred, and no partnership exists.	Phase 4

Table 2. Open gaps and the phase that closes each one.

3. Market Landscape and Positioning

The buy-side problem is well documented. WHO GLASS reports that laboratory capacity and bloodstream-infection surveillance remain the binding constraints across participating countries, and published GLASS evaluations repeatedly flag data-quality and comparability as the first-order issue. Surveillance budgets are growing roughly six percent per year, but the analytics layer between instruments and decisions is dominated by closed vendor ecosystems. The result is an unoccupied niche: an independent, license-clean audit layer that any party can run against any dataset or model, before acting on it.

3.1 Who pays, and for what

- **Public-health agencies and reference labs** - need data-quality evidence before acting on surveillance data or submitting to GLASS-style programmes; program-funded budgets.
- **Pharmaceutical and biotech AMR programs** - need dataset due diligence before licensing data, and a resistance lens when prioritizing targets and compounds (the Vecura-style workflow).
- **Academic ML groups** - need leakage audits and DOME/TRIPOD+AI compliance to pass peer review; price-sensitive, they form the free-tier funnel.
- **Diagnostics companies** - need validation evidence packs during development, long before regulatory submission.

3.2 Competitive whitespace

Player	What it sells	What it does not do
ResFinder / ABRicate	WGS AMR gene detection; best-in-class balanced accuracy around 0.80 in a 2026 systematic comparison.	No dataset-quality audit, no leakage quantification, no uncertainty guarantees.
Mykrobe	Offline species ID and resistance prediction.	Point predictions; no conformal sets; no buy-side due diligence.
DeepARG	Deep-learning AMR gene classification.	Black-box; no publication-compliance tooling.
AMRFinderPlus	Public-domain determinant detection (used by LACUNA for features).	Detection only; no validation layer.
bioMerieux / BD / Thermo Fisher	AST instruments plus lab software ecosystems.	Closed ecosystems; no independent audit of external datasets or models.
CARD / RGI	Curated resistance database.	License incompatible with an Apache-2.0 core; excluded from LACUNA by design.

Table 3. Competitive landscape: prediction tools and instrument vendors leave the audit layer unoccupied.

3.3 Unique problem-solving capacities

#	Capacity	Why competitors do not have it
U1	Leakage audit: lineage-aware splits S0-S3, clonal dedup (max 3 per clone-cluster x BioProject), measured lineage leakage.	Requires the split methodology LACUNA implements; point-metric tools never quantify leakage.
U2	Conformal prediction sets (Mondrian + marginal, 0.90 target coverage) with cluster-bootstrap intervals.	Off-the-shelf predictors are overconfident; guarantees require calibration carve-outs and conformal machinery.
U3	Bring-your-own breakpoints (EUCAST/CLSI upload, never bundled).	Vendors lock jurisdictions; bundling creates versioning and licensing liability.
U4	Publication compliance automation (DOME + TRIPOD+AI checklists, banned-claims audit).	No competitor targets the peer-review bottleneck.
U5	License-clean Apache-2.0 core.	Most AMR stacks mix incompatible licenses.
U6	Agent-platform native: OpenAPI 3.1 + MCP tool server + signed webhooks.	Bioinformatics tooling is CLI-first; LACUNA is callable inside agentic workflows.

Table 4. The six defensible differentiators shipped in v0.4.0.

Positioning statement: for every organization that must trust an AMR dataset or model before acting on it, LACUNA-AMR is the independent audit layer that measures leakage, enforces data gates, bounds prediction uncertainty, and generates publication-ready compliance evidence - unlike prediction tools and instrument vendors, whose incentives are tied to their own outputs.

4. The Vecura Integration

Vecura is an agentic AI platform for life-science discovery built by Nanyang Biologics (NYB.AI), Singapore. Vecura 2.0 (June 2026) moved beyond tool access to workflow execution: research teams connect external APIs, models and analysis tools into agent-run discovery pipelines spanning target identification, screening and translational planning. That architecture defines the integration surface LACUNA has implemented: a documented REST contract, an MCP tool server, and signed webhooks - the three mechanisms such platforms consume natively.

The value loop for a Vecura workflow user is the resistance lens. When an agent workflow evaluates natural-product hits or prioritizes targets, LACUNA contributes the audit and uncertainty layer: it certifies the AMR datasets feeding the evaluation, returns conformal resistance predictions instead of overconfident point calls, and flags determinants that are conserved versus divergent across resistant lineages. Downstream prioritization is then built on measured, not assumed, data quality.

Integration component	Shipped in v0.4.0	Remaining for general availability
OpenAPI 3.1 contract	GET /api/openapi.json covering all /api/v1 routes with RUO and credit-cost extensions.	Publish alongside API docs on the marketing site.

Integration component	Shipped in v0.4.0	Remaining for general availability
MCP tool server	POST /api/mcp: initialize, tools/list (5 tools), tools/call with the same auth and metering as REST.	Joint round-trip test with the actual platform; version pinning.
Signed webhooks	HMAC-SHA256 outbound events (audit/predict/validate/report completed) with honest delivery log; inbound receiver with timing-safe verification.	Retry policy and replay endpoint for production use.
Console connector	Integrations section: endpoint configuration, event subscriptions, secret rotation, connection test, event log, usage metering.	Per-organization credentials once multi-tenancy lands.

Table 5. Vecura-ready integration: shipped versus remaining.

> **Integrity boundary**
> The connector is built to published patterns (REST + OpenAPI + JSON-RPC MCP + HMAC webhooks) and is labelled Vecura-ready. LACUNA-AMR claims no partnership with, or inside knowledge of, Nanyang Biologics or NYB.AI; the platform simply consumes standard integration surfaces. Marketing copy must preserve this distinction until a joint test and any agreement exist.

5. Phase 1 - Production Foundation (Weeks 1-6)

Phase 1 turns the verified v0.4.0 demo into a deployable commercial service. The engineering is deliberately unglamorous: identity, billing, database, deployment and observability. Nothing in this phase touches the scientific core, which keeps the open-core boundary clean - the Apache-2.0 package remains the reference implementation, and the commercial layer only orchestrates it.

Workstream	Deliverable	Exit criterion
Multi-tenant auth	Email and password or SSO sign-in; organizations, memberships, roles; API keys scoped to organizations.	Two organizations can coexist with isolated data.
Billing	Stripe integration: three tiers (community free, lab subscription, enterprise per-engagement) mapped to existing scopes and credits; invoices and dunning.	A test subscription grants and revokes entitlements end-to-end.
Database	Postgres migration with Prisma; connection pooling; nightly backups; restore drill documented.	Migration runbook executed once against production-shaped data.
Deployment	Containerized deploy on a managed platform; environment promotion (staging, production); secrets management; uptime monitoring.	One-command deploy from a clean checkout.
Observability	Structured logs, error tracking, uptime dashboard; webhook delivery metrics.	Alert on error-rate and webhook-failure thresholds.
Legal entity	Company or equivalent registered; terms of service, privacy policy, and data-processing agreement templates.	Stripe can legally accept charges.

Table 6. Phase 1 workstreams. Effort estimate: two engineers for six weeks.

6. Phase 2 - Scientific Credibility (Weeks 4-12)

Credibility is the product. The pilot is honest but exploratory (240 isolates against a data gate requiring 800 per drug), and the market will ask for the full-scale run and an external benchmark before any enterprise engagement. Phase 2 executes the full-scale cohort on adequate hardware (cloud burst or a partner machine - the pipeline is Snakemake and already specified), then turns the resulting evidence into a preprint and two design-partner validations.

Workstream	Deliverable	Exit criterion
Full-scale DG cohort	Execute the frozen pipeline at data-gate scale (at least 800 MIC isolates per drug, 100 R and 100 S, 8 lineages with 25 each, 5 BioProjects) on rented hardware.	All data gates pass on real data; results recorded fold-for-fold reproducibly.
External benchmark	Head-to-head audit comparison against ResFinder/ABRicate-class outputs on shared isolates; leakage and coverage reported for every tool.	Benchmark table published with code and data pointers.
Preprint	Dual-journal manuscript (conservative generic structure; venue formatting deferred to submission) on the audit methodology and full-scale results.	Posted to a preprint server with DOIs re-verified at runtime.
Design partners	Two pilot organizations (one reference lab, one AMR research group) running weekly audits; feedback loop into the backlog.	Weekly usage by both partners for four consecutive weeks.

Table 7. Phase 2 workstreams. Effort: one scientist plus one engineer, weeks 4-12, overlapping Phase 1.

7. Phase 3 - Regulatory and Compliance (Weeks 8-20)

LACUNA-AMR launches as research-use-only, and every claim in the product must keep that true. Phase 3 makes the posture auditable instead of aspirational: the banned-claims auditor already ships as a feature, and here it runs against all marketing surfaces as a release gate. The EU AI Act applies high-risk obligations to new products from August 2026 and to existing CE-marked products from August 2027; as long as LACUNA makes no diagnostic or clinical claims it stays outside medical-device classification, and the documented boundary conditions (RUO notices, user-supplied breakpoints, no interpretive outputs) preserve that position. The phase also covers data protection (GDPR and India DPDP) and the groundwork for SOC 2 Type I.

Workstream	Deliverable	Exit criterion
Claims audit	Banned-claims scan wired into CI over console copy, API descriptions and marketing pages; zero hits enforced.	Release blocked on any positive hit.
Regulatory memo v2	Updated regulatory-risk memo covering EU AI Act, MDR/IVDR boundaries and the conditional path if diagnostic claims are ever pursued.	Reviewed by external counsel.

Workstream	Deliverable	Exit criterion
Data protection	Records of processing, DPAs with design partners, data-retention policy for uploaded datasets.	DPA signed by both Phase-2 design partners.
SOC 2 Type I groundwork	Control mapping, access reviews, incident process, vendor register.	Readiness assessment passed; Type II observation window opened.

Table 8. Phase 3 workstreams. Effort: fractional compliance lead, weeks 8-20.

8. Phase 4 - Go-To-Market (Weeks 12-24)

With foundations and credibility in place, Phase 4 turns the roadmap's pricing posture into validated numbers. Pricing today is deliberately positioned against published benchmark ranges rather than invented: community tier free, lab subscription in the published clinical-genomics SaaS band, enterprise per engagement. Two design-partner cycles test willingness to pay for the three service SKUs (dataset audit report, validation audit, unexplained-resistance classification), and the Vecura integration reaches general availability through a joint test.

Workstream	Deliverable	Exit criterion
Pricing validation	Two design-partner pricing cycles; conversion and usage instrumented.	Pricing set from evidence; grandfathering policy documented.
Vecura GA	Joint end-to-end MCP round-trip test; integration guide; listing or reference on both sides' documentation.	A Vecura workflow invokes LACUNA tools in a recorded session.
Content engine	DOME and TRIPOD+AI explainers, leakage-audit methodology posts, benchmark write-ups; conference abstracts (ECCMID, ASM).	Two abstracts submitted; four posts published.
Self-serve funnel	Onboarding flow: signup, key issue, first audit call in under 15 minutes; in-app examples.	Median time-to-first-successful-call under 15 minutes.
SKU launch	The three service SKUs open for purchase with scoped statements of work.	First paid engagement signed and delivered.

Table 9. Phase 4 workstreams. Effort: founder-led sales plus one engineer, weeks 12-24.

9. Phase 5 - Scale (Months 6-18)

Phase 5 expands only after the earlier exit criteria hold. Enterprise buyers will ask for SSO, audit exports and SLAs - all natural extensions of the existing hash-chained audit trail and API-key scopes. The scientific surface expands along the pre-registered fallback chain (more organisms and drugs under the same data gates), and the integration surface grows from one platform pattern to an ecosystem: any OpenAPI-plus-MCP consumer can register LACUNA tools.

- **Enterprise readiness:** SSO/SAML, role-based access, per-organization webhooks and keys, SLA tiers, on-prem runner for air-gapped reference labs.

- **Organism and drug expansion:** Klebsiella pneumoniae and Acinetobacter baumannii under the same data gates; colistin re-entry only with broth-microdilution-only labelling.
- **API ecosystem:** SDKs (Python and TypeScript) generated from the OpenAPI spec; a partner registry for workflow platforms.
- **Certified audits marketplace:** accredited third parties deliver the service SKUs on the platform, with LACUNA supplying the tooling and the evidence chain.
- **Optional SaMD path:** only if a payer-funded clinical use case justifies MDR/IVD classification; the RUO posture is otherwise permanent.

10. KPIs, Risks and Decision Points

Phase	Exit KPI	Target
1 Foundation	Deploy success; two organizations coexist; test subscription round-trip.	All three demonstrated
2 Credibility	Full-scale gates pass; preprint posted; design partners weekly-active.	4 consecutive weeks x 2 partners
3 Compliance	Banned-claims CI gate; counsel-reviewed memo; DPAs signed.	Zero hits; 2 DPAs
4 GTM	Time-to-first-call; Vecura joint round-trip; first paid SKU.	<15 min; recorded; signed
5 Scale	Enterprise features live; second organism shipped; SDK downloads.	Quarterly review

Table 10. Phase exit KPIs.

10.1 Risk register

Risk	Likelihood	Impact	Mitigation
Regulatory reclassification (a claim drifts into diagnostic territory).	Low	High	CI banned-claims gate; counsel-reviewed memo; RUO notices on every predictive surface.
Instrument vendor ships a comparable audit layer.	Medium	High	Open-core license cleanliness and independence are the differentiators vendors cannot copy without conflict; accelerate Phase 2 benchmark.
Full-scale run fails a data gate (no organism reaches 800 per drug).	Medium	Medium	Pre-registered fallback chain; report honestly either way - the audit tool's credibility survives a FAIL verdict; the pipeline's does not have to.
Vecura joint test never happens or integration diverges.	Medium	Medium	Connector is pattern-based, not platform-specific; any OpenAPI/MCP consumer substitutes; no partnership claims to retract.
Pricing below viability for the service SKUs.	Medium	Medium	Two design-partner pricing cycles before general launch; service SKUs scoped per drug and species.

Table 11. Risk register with mitigations.

10.2 Decision points

Two scheduled decisions keep the roadmap honest. First, at the end of Phase 2: if the full-scale cohort cannot pass data gates on any organism, the product pivots from validated-predictions to audit-only positioning (the gates, leakage and compliance layers stand on their own). Second, at the end of Phase 4: if two pricing cycles produce no paid engagement and weekly usage is flat, the service SKUs go service-led (consulting-led delivery using the platform) rather than self-serve, and self-serve becomes the free funnel only. Both decisions are reversible; neither gambles the scientific reputation.

11. Immediate Next Actions

The following checklist is executable this week with no new hires and no spend beyond a small cloud instance for the full-scale rehearsal.

#	Action	Owner	Output
1	Provision staging environment; deploy the console from a clean checkout; smoke-test the 14 sections.	Engineering	Staging URL + deploy runbook
2	Open the legal-entity and Stripe checklist; draft terms of service and privacy policy.	Founder	Filed applications
3	Rent an 8-core cloud instance; rehearse the full-scale pipeline at 25% DG scale to size the real run.	Science	Costed full-scale plan
4	Contact two candidate design partners (one reference lab, one AMR group) with the audit console demo.	Founder	Two letters of intent
5	Request a Vecura joint integration test via public channels; prepare the MCP round-trip script.	Founder	Test session scheduled or documented attempt
6	Wire the banned-claims scanner into CI over all shipped copy.	Engineering	Green CI gate
7	Publish the API documentation page generated from the OpenAPI spec.	Engineering	Public docs URL
8	Write the Phase-2 benchmark protocol and pre-register it.	Science	Pre-registration record

Table 12. This week's actions, each with an owner and a concrete output.

Sources

All external figures in this roadmap carry a live source; project-internal claims carry a repository artifact. Market figures were retrieved on 2026-11-09; pricing posture follows deviation D-024 (benchmark-cited positioning, not a price list).

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